# Why Do Employees Leave? What the IBM HR Attrition Data Says, and What I Got Wrong

I am a developer and a beginner in data analysis. I had just finished an Excel course, and I wanted to learn by doing something real. So I picked one question:

### Which things are connected to employees leaving a company?

This is the story of how I tried to answer it: what I found, what confused me, and one mistake that taught me more than the results.

> **Please note:** the IBM HR dataset is fictional, created by IBM data scientists. What I found describes this dataset only, and everything here is an **association, not a cause**.

## Chapter 1: Meeting the data

The dataset has 1,470 employees. 237 of them left, which is 16.1%. I sorted the variables into three types, because each type needs different tools: *scale* (numbers like income and years), *ordinal* (ordered levels like satisfaction), and *nominal* (labels like job role). I removed three columns that never change (EmployeeCount, StandardHours, Over18), since they cannot explain anything.

Then I started in Excel, where I felt comfortable, and asked simple questions.

**Does overtime matter?** It was the first surprise. Among people who work overtime, **30.5% left**. Among people who don't, **10.4%**.

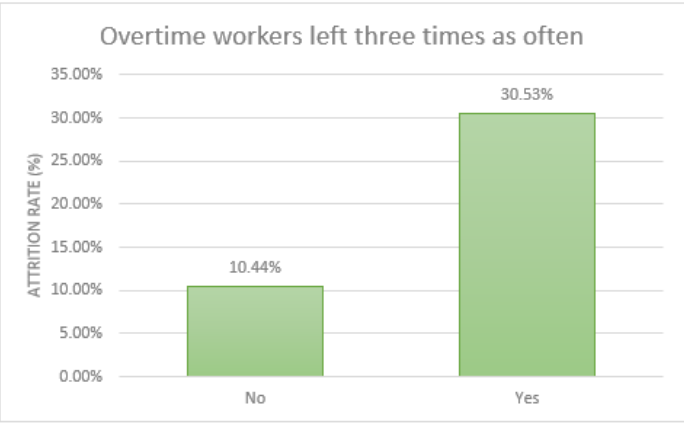



***Figure 1**: Attrition rate by overtime status*

**Does seniority matter?** Entry-level employees left the most (**26.3%**). At the top levels the rate was only 4.7% and 7.3%.

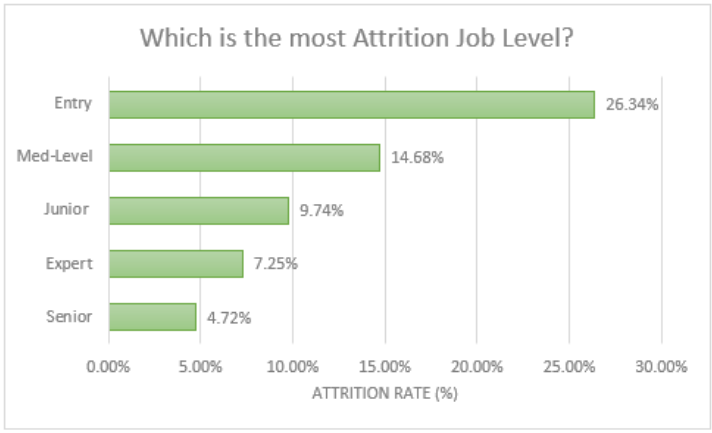



***Figure 2:** Attrition rate by job level*

**Which roles are at risk? Sales Representatives** stood out at about 40%, with Laboratory Technicians and Human Resources next. In the chart, bars show how many people are in each role and the line shows the attrition rate.

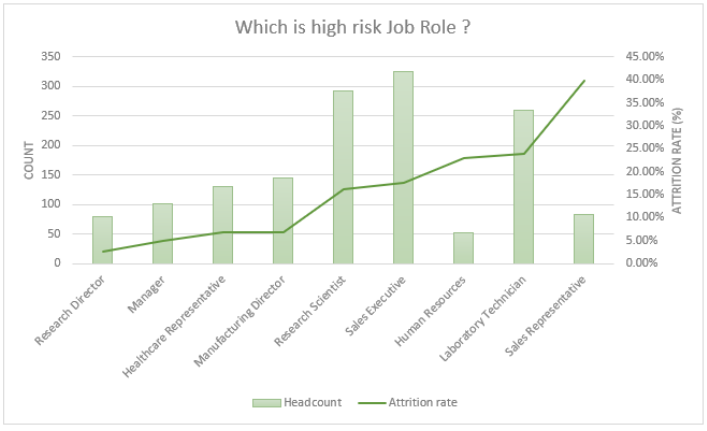



***Figure 3**: Headcount (bars) and attrition rate (line) by job role*

**Does travel matter?** Yes: 8.0% for non-travellers, 15.0% for people who travel rarely, and 24.9% for people who travel frequently.

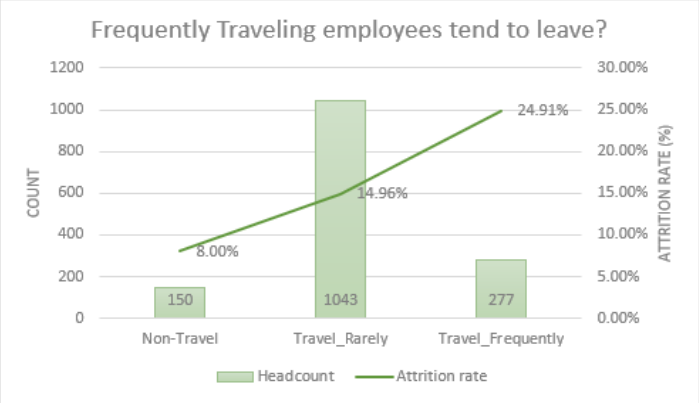



***Figure 4:*** *Headcount (bars) and attrition rate (line) by business travel*

**Does satisfaction matter?** Attrition was 22.8% at the lowest job satisfaction and 11.3% at the highest. But the middle levels were almost the same (about 16.5%), so it was not a perfectly steady slope. I remembered this later.

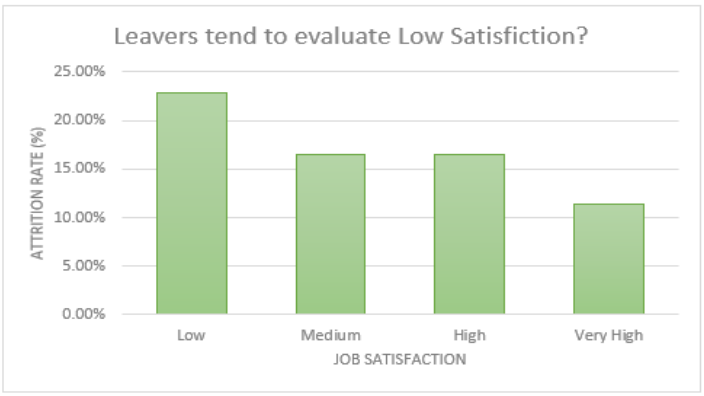



***Figure 5:*** *Attrition rate by job satisfaction level*

I also looked at age, department and salary. Younger employees left more often (the youngest group has only 28 people, so its 57% is not reliable), and Sales and HR had higher rates than Research & Development. When I compared the salaries of leavers and stayers, they were similar within each job level, which told me that pay mostly follows job level.

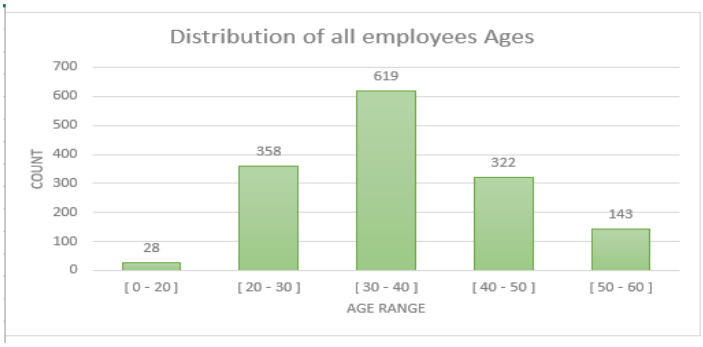



***Figure 6:*** *Age distribution of all employees*

The distribution of leavers employees per age is shown below

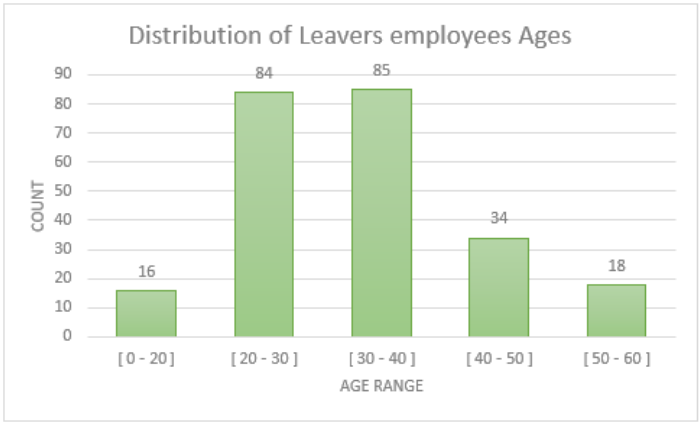



***Figure 7:*** *Age distribution of employees who left*

Attrition rate for each age range

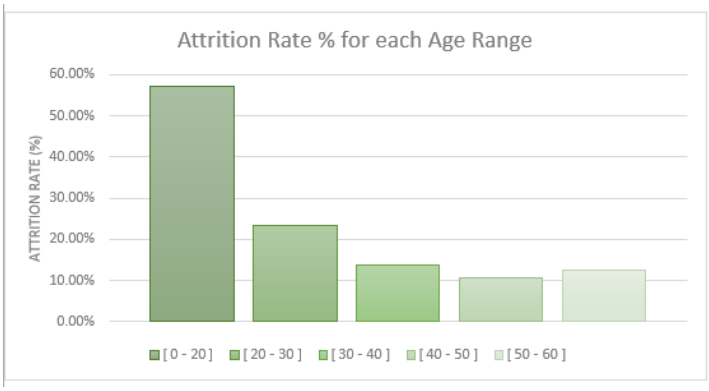



***Figure 8:*** *Attrition rate by age range*

Next, we'll look at the distribution of older vs. younger generations within organizations and compare the two groups' ages to see how that distribution appears

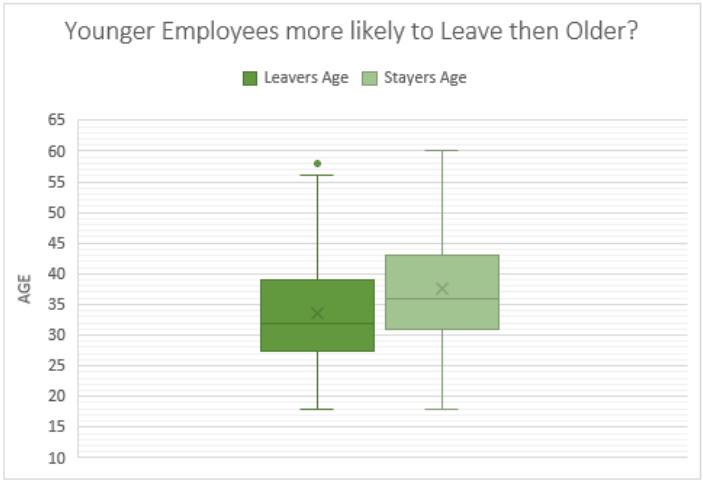



***Figure 9:*** *Age box plot: leavers vs stayers*

Then, we look at top risk department by compare and attrition rate of each department with the headcount, which was **Human Resources and Sales.**

Attrition rate for each department

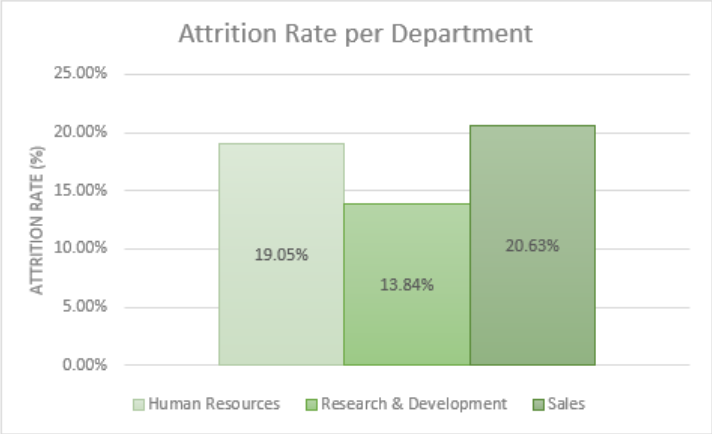



***Figure 10:*** *Attrition rate by department*

The distribution of employees per department is shown below.

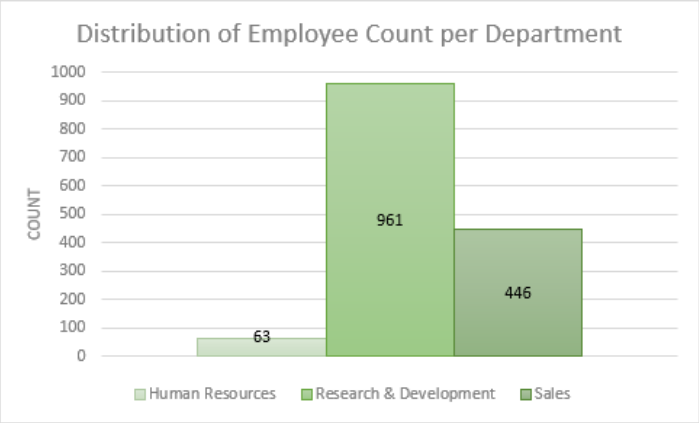



***Figure 11****: Headcount by department*

Compared the salaries of leavers and stayers, for each job level and department

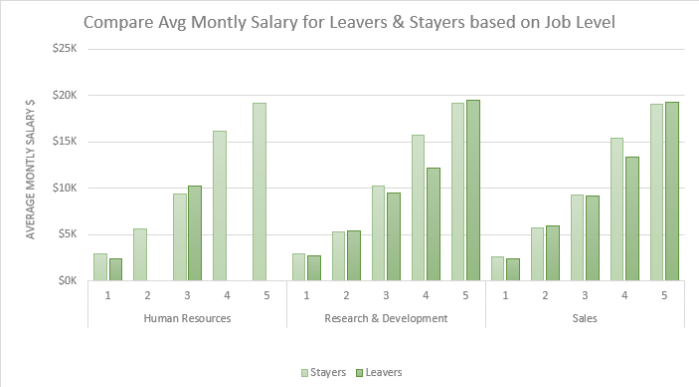



***Figure 12:*** *Average monthly salary by department and job level, stayers vs leavers*

# Chapter 2: Is it real, or just chance?

Charts show patterns, but patterns can be luck. I moved to SPSS and tested each variable against Attrition. The important lesson: the right test depends on the type of data, and a tiny p-value is not enough, because with 1,470 rows almost everything is "significant". So I always looked at the **effect size** too.

| Question | Test | Result |
|---|---|---|
| **Job role and attrition** | Chi-square | $p < .001$, Cramér's V = 0.242 (small to medium) |
| **Overtime and attrition** | Chi-square | $\chi^2(1) = 89.04$, $p < .001$, Cramér's V = 0.246 (small to medium) |
| **Business travel and attrition** | Chi-square | $\chi^2(2) = 24.18$, $p < .001$, Cramér's V = 0.128 (small) |
| **Department and attrition** | Chi-square | $\chi^2(2) = 10.80$, $p = .005$, Cramér's V = 0.086 (very small) |
| **Job level and attrition** | Mann-Whitney U | $Z = -7.30$, $r \approx 0.19$: leavers sit at lower levels |
| **Job involvement** | Mann-Whitney U | $Z = -4.58$, $r \approx 0.12$ (small) |
| **Job satisfaction** | Mann-Whitney U | $Z = -3.95$, $r \approx 0.10$ (small) |

Overtime turned out to have the largest effect size (V = 0.246), almost the same as job role (0.242), while department was the weakest (V = 0.086). Here is the SPSS output for the ordinal variables and for the other three crosstabs:

**Ranks**

| | Attr... | N | Mean Rank | Sum of Ranks |
|---|---|---|---|---|
| JobLevel | No | 1233 | 769.06 | 948252.50 |
| | Yes | 237 | 560.90 | 132932.50 |
| | Total | 1470 | | |

**Test Statistics[a]**

| | JobLevel |
|---|---|
| Mann-Whitney U | 1.047E5 |
| Wilcoxon W | 1.329E5 |
| Z | -7.296 |
| Asymp. Sig. (2-tailed) | .000 |

a. Grouping Variable: Attrition

***Figure 13:*** *SPSS Mann-Whitney U: job level by attrition*

**Case Processing Summary**

| | Cases | | | | | |
|---|---|---|---|---|---|---|
| | Valid | | Missing | | Total | |
| | N | Percent | N | Percent | N | Percent |
| JobRole * Attrition | 1470 | 100.0% | 0 | .0% | 1470 | 100.0% |

**JobRole * Attrition Crosstabulation**

Count

| | | Attrition | | |
|---|---|---|---|---|
| | | No | Yes | Total |
| JobRole | Healthcare Representative | 122 | 9 | 131 |
| | Human Resources | 40 | 12 | 52 |
| | Laboratory Technician | 197 | 62 | 259 |
| | Manager | 97 | 5 | 102 |
| | Manufacturing Director | 135 | 10 | 145 |
| | Research Director | 78 | 2 | 80 |
| | Research Scientist | 245 | 47 | 292 |
| | Sales Executive | 269 | 57 | 326 |
| | Sales Representative | 50 | 33 | 83 |
| Total | | 1233 | 237 | 1470 |

**Chi-Square Tests**

| | Value | df | Asymp. Sig. (2-sided) |
|---|---|---|---|
| Pearson Chi-Square | 86.190[a] | 8 | .000 |
| Likelihood Ratio | 88.909 | 8 | .000 |
| Linear-by-Linear Association | 6.624 | 1 | .010 |
| N of Valid Cases | 1470 | | |

a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 8.38.

**Symmetric Measures**

| | | Value | Approx. Sig. |
|---|---|---|---|
| Nominal by Nominal | Phi | .242 | .000 |
| | Cramer's V | .242 | .000 |
| N of Valid Cases | | 1470 | |

***Figure 14:*** *SPSS crosstab: job role by attrition (chi-square & Cramér's V)*

## Mann-Whitney

**Ranks**

| | Attr... | N | Mean Rank | Sum of Ranks |
|---|---|---|---|---|
| JobSatisfaction | No | 1233 | 753.95 | 929619.50 |
| | Yes | 237 | 639.52 | 151565.50 |
| | Total | 1470 | | |
| JobInvolvement | No | 1233 | 755.09 | 931025.00 |
| | Yes | 237 | 633.59 | 150160.00 |
| | Total | 1470 | | |

**Test Statistics[a]**

| | Job Satisfaction | Job Involvement |
|---|---|---|
| Mann-Whitney U | 123362.500 | 121957.000 |
| Wilcoxon W | 151565.500 | 150160.000 |
| Z | -3.946 | -4.580 |
| Asymp. Sig. (2-tailed) | .000 | .000 |

a. Grouping Variable: Attrition

***Figure 15:*** *Mann-Whitney U: job satisfaction and job involvement by attrition*

**Case Processing Summary**

| | Cases | | | | | |
|---|---|---|---|---|---|---|
| | Valid | | Missing | | Total | |
| | N | Percent | N | Percent | N | Percent |
| OverTime * Attrition | 1470 | 100.0% | 0 | .0% | 1470 | 100.0% |

**OverTime * Attrition Crosstabulation**

| | | | Attrition | | |
|---|---|---|---|---|---|
| | | | No | Yes | Total |
| OverTime | No | Count | 944 | 110 | 1054 |
| | | Expected Count | 884.1 | 169.9 | 1054.0 |
| | Yes | Count | 289 | 127 | 416 |
| | | Expected Count | 348.9 | 67.1 | 416.0 |
| Total | | Count | 1233 | 237 | 1470 |
| | | Expected Count | 1233.0 | 237.0 | 1470.0 |

**Chi-Square Tests**

| | Value | df | Asymp. Sig. (2-sided) | Exact Sig. (2-sided) | Exact Sig. (1-sided) |
|---|---|---|---|---|---|
| Pearson Chi-Square | 89.044[a] | 1 | .000 | | |
| Continuity Correction[b] | 87.564 | 1 | .000 | | |
| Likelihood Ratio | 81.402 | 1 | .000 | | |
| Fisher's Exact Test | | | | .000 | .000 |
| Linear-by-Linear Association | 88.983 | 1 | .000 | | |
| N of Valid Cases[b] | 1470 | | | | |

a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 67.07.

b. Computed only for a 2x2 table

**Symmetric Measures**

| | | Value | Approx. Sig. |
|---|---|---|---|
| Nominal by Nominal | Phi | .246 | .000 |
| | Cramer's V | .246 | .000 |
| N of Valid Cases | | 1470 | |

***Figure 16:*** *Crosstab. overtime by attrition*

Then I looked at how the numeric variables relate to each other and to Attrition.

**Pearson Correlations**

| | Attrition | DistanceFromHome | MonthlyIncome | NumCompaniesWorked | PercentSalaryHike | TotalWorkingYears | TrainingTimesLastYear | YearsAtCompany | YearsInCurrentRole | YearsSinceLastPromotion | YearsWithCurrManager |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Attrition | 1 | 0.079 | -0.198 | 0.031 | -0.024 | -0.199 | -0.052 | -0.19 | -0.181 | -0.053 | -0.175 |
| DistanceFromHome | 0.079 | 1 | -0.017 | -0.029 | 0.04 | 0.005 | -0.037 | 0.01 | 0.019 | 0.01 | 0.014 |
| MonthlyIncome | -0.198 | -0.017 | 1 | 0.15 | -0.027 | 0.773 | -0.022 | 0.514 | 0.364 | 0.345 | 0.344 |
| NumCompaniesWorked | 0.031 | -0.029 | 0.15 | 1 | -0.01 | 0.238 | -0.066 | -0.118 | -0.091 | -0.037 | -0.11 |
| PercentSalaryHike | -0.024 | 0.04 | -0.027 | -0.01 | 1 | -0.021 | -0.005 | -0.036 | -0.002 | -0.022 | -0.012 |
| TotalWorkingYears | -0.199 | 0.005 | 0.773 | 0.238 | -0.021 | 1 | -0.036 | 0.628 | 0.46 | 0.405 | 0.459 |
| TrainingTimesLastYear | -0.052 | -0.037 | -0.022 | -0.066 | -0.005 | -0.036 | 1 | 0.004 | -0.006 | -0.002 | -0.004 |
| YearsAtCompany | -0.19 | 0.01 | 0.514 | -0.118 | -0.036 | 0.628 | 0.004 | 1 | 0.759 | 0.618 | 0.769 |
| YearsInCurrentRole | -0.181 | 0.019 | 0.364 | -0.091 | -0.002 | 0.46 | -0.006 | 0.759 | 1 | 0.548 | 0.714 |
| YearsSinceLastPromotion | -0.053 | 0.01 | 0.345 | -0.037 | -0.022 | 0.405 | -0.002 | 0.618 | 0.548 | 1 | 0.51 |
| YearsWithCurrManager | -0.175 | 0.014 | 0.344 | -0.11 | -0.012 | 0.459 | -0.004 | 0.769 | 0.714 | 0.51 | 1 |

***Figure 17:*** *Pearson correlations: scale variables and Attrition*

**Spearman Correlation**

| | DistanceFromHome | MonthlyIncome | NumCompaniesWorked | PercentSalaryHike | TotalWorkingYears | TrainingTimesLastYear | YearsAtCompany | YearsInCurrentRole | YearsSinceLastPromotion | YearsWithCurrManager |
|---|---|---|---|---|---|---|---|---|---|---|
| DistanceFromHome | 1 | 0.003 | -0.01 | 0.03 | -0.003 | -0.025 | 0.011 | 0.014 | -0.005 | 0.004 |
| MonthlyIncome | 0.003 | 1 | 0.19 | -0.034 | 0.71 | -0.035 | 0.464 | 0.395 | 0.265 | 0.365 |
| NumCompaniesWorked | -0.01 | 0.19 | 1 | 0 | 0.315 | -0.047 | -0.171 | -0.128 | -0.067 | -0.144 |
| PercentSalaryHike | 0.03 | -0.034 | 0 | 1 | -0.026 | -0.004 | -0.054 | -0.026 | -0.055 | -0.026 |
| TotalWorkingYears | -0.003 | 0.71 | 0.315 | -0.026 | 1 | -0.014 | 0.594 | 0.493 | 0.335 | 0.495 |
| TrainingTimesLastYear | -0.025 | -0.035 | -0.047 | -0.004 | -0.014 | 1 | 0.001 | 0.005 | 0.01 | -0.012 |
| YearsAtCompany | 0.011 | 0.464 | -0.171 | -0.054 | 0.594 | 0.001 | 1 | 0.854 | 0.52 | 0.843 |
| YearsInCurrentRole | 0.014 | 0.395 | -0.128 | -0.026 | 0.493 | 0.005 | 0.854 | 1 | 0.506 | 0.725 |
| YearsSinceLastPromotion | -0.005 | 0.265 | -0.067 | -0.055 | 0.335 | 0.01 | 0.52 | 0.506 | 1 | 0.467 |
| YearsWithCurrManager | 0.004 | 0.365 | -0.144 | -0.026 | 0.495 | -0.012 | 0.843 | 0.725 | 0.467 | 1 |

***Figure 18:*** *Spearman correlations: scale variables*

What the matrices told me:

- Nothing numeric is strongly linked to Attrition. The biggest correlations are about −0.20 (TotalWorkingYears, MonthlyIncome, YearsAtCompany).
- Many variables are almost copies of each other: JobLevel and MonthlyIncome (0.92), and the three tenure variables (0.72 to 0.85). The final model uses JobLevel and YearsWithCurrManager but not MonthlyIncome, TotalWorkingYears or YearsAtCompany, which overlap with them.
- Pearson and Spearman agree for most pairs, but not for the tenure variables (0.76 vs 0.85), which means the relationship is steady but not a straight line.
- The satisfaction variables are almost unrelated to each other (|r| < 0.06), so each carries its own information.

# Chapter 3: Can I trust my tools?

I also had a question about money: do leavers and stayers earn different amounts once job level is taken into account? To test it I planned a two-way ANOVA (monthly income by attrition and job level). But a test is only trustworthy when its assumptions hold, and ANOVA assumes that the model's *residuals* are roughly normal with similar spread across groups. So before using it, I checked. Raw monthly income is clearly not normal: it has a floor and a cap, so its Q-Q plot bends into an S-shape. But the *residuals* of my ANOVA model were close to normal, because job level explains most of the income pattern.

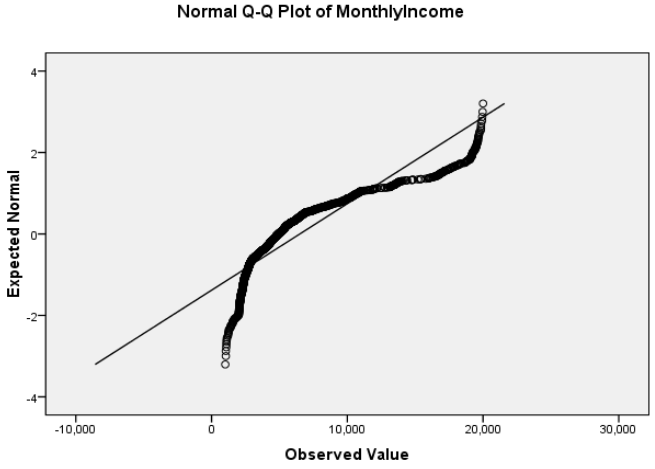



***Figure 20:*** *Q-Q plot of raw MonthlyIncome*

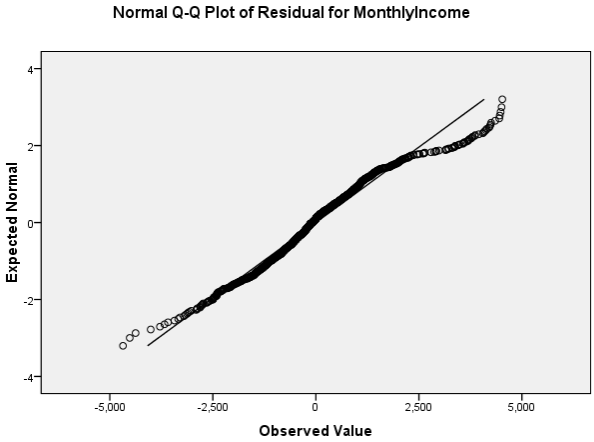



***Figure 21:*** *Q-Q plot of the ANOVA residuals*

The histogram agrees with the Q-Q plot: bell-shaped, centred on 0, with a slightly heavier right tail.

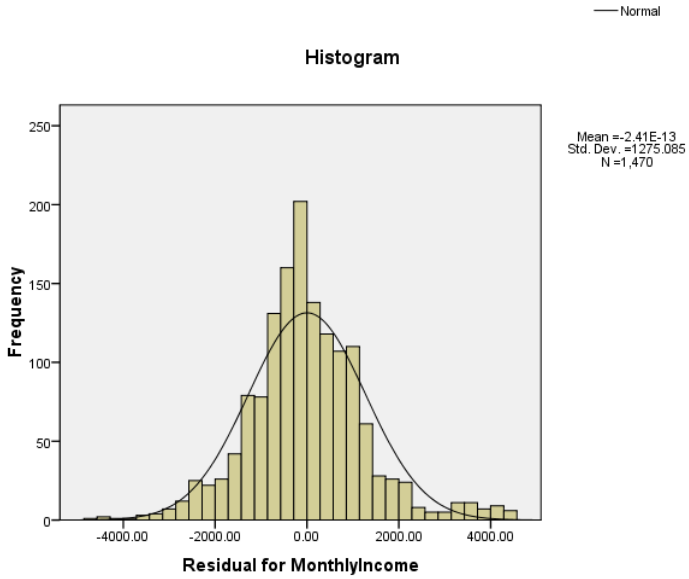



***Figure 22:*** *Histogram of the ANOVA residuals*

The variance was not equal across job levels (narrow at the two ends, wide in the middle), which I learned to read from the residual plot.

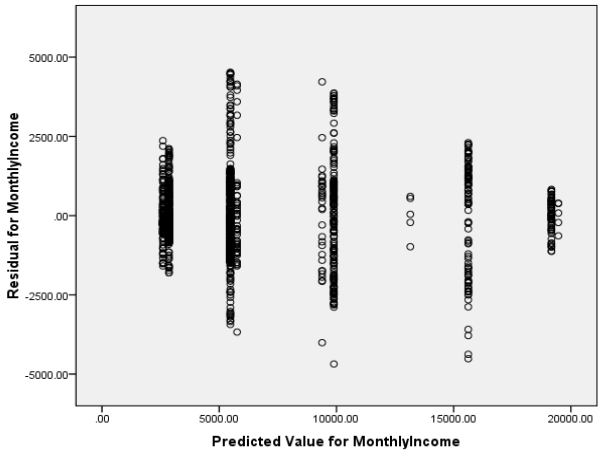



***Figure 23:*** *Residuals vs predicted values of MonthlyIncome*

Then the test itself. **Job level explains most of the income** (partial $\eta^2 = .81$), while **attrition alone explains almost nothing** (partial $\eta^2 = .006$). There is a small interaction (partial $\eta^2 = .016$): leavers and stayers earn about the same at levels 1, 2, 3 and 5, and the only noticeable gap is at level 4, where leavers earn about 2,470 less. Levels 4 and 5 have only 5 leavers each, so I treat that gap carefully. Levene's test showed unequal variances, so I confirmed the interaction with robust (HC3) standard errors: the level-4 coefficient is 2,763 (95% CI 1,912 to 3,614, $p < .001$). In short, money follows job level, which fits the 0.92 correlation between the two and explains why I did not need income as a separate predictor.

**Levene's Test of Equality of Error Variances[a,b]**

| | | Levene Statistic | df1 | df2 | Sig. |
|---|---|---|---|---|---|
| MonthlyIncome | Based on Mean | 36.617 | 9 | 1460 | <.001 |
| | Based on Median | 30.466 | 9 | 1460 | <.001 |
| | Based on Median and with adjusted df | 30.466 | 9 | 1065.118 | <.001 |
| | Based on trimmed mean | 35.794 | 9 | 1460 | <.001 |

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.

a. Dependent variable: MonthlyIncome

b. Design: Intercept + Attrition + JobLevel + Attrition * JobLevel

**Tests of Between-Subjects Effects**

Dependent Variable: MonthlyIncome

| Source | Type III Sum of Squares | df | Mean Square | F | Sig. | Partial Eta Squared |
|---|---|---|---|---|---|---|
| Corrected Model | 3.017E+10[a] | 9 | 3352423603 | 2049.328 | .000 | .927 |
| Intercept | 2.168E+10 | 1 | 2.168E+10 | 13251.015 | .000 | .901 |
| Attrition | 14248857.20 | 1 | 14248857.20 | 8.710 | .003 | .006 |
| JobLevel | 1.007E+10 | 4 | 2516582986 | 1538.381 | .000 | .808 |
| Attrition * JobLevel | 39562989.72 | 4 | 9890747.429 | 6.046 | <.001 | .016 |
| Error | 2388362609 | 1460 | 1635864.801 | | | |
| Total | 9.472E+10 | 1470 | | | | |
| Corrected Total | 3.256E+10 | 1469 | | | | |

a. R Squared = .927 (Adjusted R Squared = .926)

***Figure 24:*** *Two-way ANOVA: Levene's test and tests of between-subjects effects*

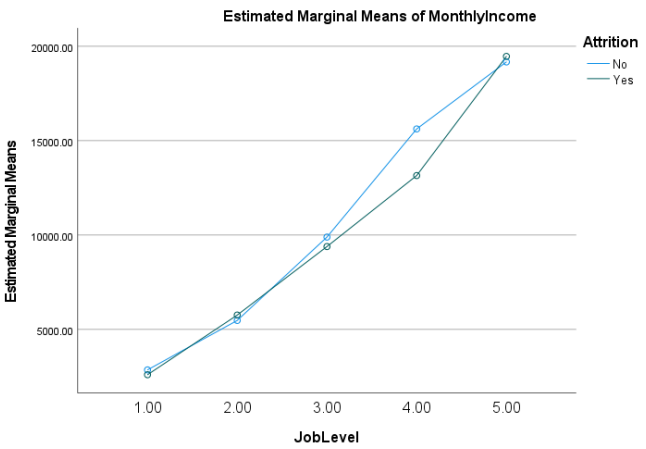



***Figure 25:*** *Profile plot: estimated mean income by job level and attrition*

**Parameter Estimates with Robust Standard Errors**

Dependent Variable: MonthlyIncome

| Parameter | B | Robust Std. Error[a] | t | Sig. | 95% Confidence Interval Lower Bound | 95% Confidence Interval Upper Bound | Partial Eta Squared |
|---|---|---|---|---|---|---|---|
| Intercept | 19463.800 | 218.642 | 89.021 | .000 | 19034.914 | 19892.686 | .844 |
| [Attrition=.00] | -293.222 | 228.055 | -1.286 | .199 | -740.573 | 154.129 | .001 |
| [Attrition=1.00] | 0[b] | . | . | . | . | . | . |
| [JobLevel=1.00] | -16865.674 | 227.358 | -74.181 | .000 | -17311.656 | -16419.692 | .790 |
| [JobLevel=2.00] | -13704.012 | 310.776 | -44.096 | <.001 | -14313.626 | -13094.397 | .571 |
| [JobLevel=3.00] | -10075.394 | 363.336 | -27.730 | <.001 | -10788.109 | -9362.678 | .345 |
| [JobLevel=4.00] | -6313.400 | 390.645 | -16.161 | <.001 | -7079.685 | -5547.115 | .152 |
| [JobLevel=5.00] | 0[b] | . | . | . | . | . | . |
| [Attrition=.00] * [JobLevel=1.00] | 549.504 | 239.308 | 2.296 | .022 | 80.078 | 1018.929 | .004 |
| [Attrition=.00] * [JobLevel=2.00] | 7.929 | 323.731 | .024 | .980 | -627.098 | 642.957 | .000 |
| [Attrition=.00] * [JobLevel=3.00] | 795.848 | 392.851 | 2.026 | .043 | 25.236 | 1566.460 | .003 |
| [Attrition=.00] * [JobLevel=4.00] | 2763.109 | 434.017 | 6.366 | <.001 | 1911.746 | 3614.472 | .027 |
| [Attrition=.00] * [JobLevel=5.00] | 0[b] | . | . | . | . | . | . |
| [Attrition=1.00] * [JobLevel=1.00] | 0[b] | . | . | . | . | . | . |
| [Attrition=1.00] * [JobLevel=2.00] | 0[b] | . | . | . | . | . | . |
| [Attrition=1.00] * [JobLevel=3.00] | 0[b] | . | . | . | . | . | . |
| [Attrition=1.00] * [JobLevel=4.00] | 0[b] | . | . | . | . | . | . |
| [Attrition=1.00] * [JobLevel=5.00] | 0[b] | . | . | . | . | . | . |

a. HC3 method

b. This parameter is set to zero because it is redundant.

***Figure 26:*** *Two-way ANOVA: parameter estimates with robust (HC3) standard errors*

# Chapter 4: Putting everything together

One variable at a time cannot tell me what still matters when the others are held fixed. For that I used logistic regression.

First, I prepared the data carefully:

1. **Overlap check (VIF):** all values below 1.5.
2. **Straight-line check (Box-Tidwell):** the satisfaction variables were not linear, so I treated them as categories.
3. **Merging thin categories:** I merged levels with similar odds ratios or too few people (for example JobLevel 4 with 5, EnvironmentSatisfaction levels 2 with 3, and JobSatisfaction levels 1 to 3 against 4).
4. **Dropping weak variables:** Age and RelationshipSatisfaction.

This took the model from 30 parameters to **22**, and its predictive power barely changed (about 0.01 in AUC), so the simpler model won. I split the employees into **70% for training** (1,060 people, 172 leavers) and **30% for testing** (410 people, 65 leavers).

Here is what the final model found (training set):

| Predictor | Odds ratio | 95% CI |
|---|---|---|
| **OverTime: yes vs no** | 4.99 | 3.31 to 7.53 |
| **YearsWithCurrManager (per year)** | 0.90 | 0.83 to 0.97 |
| **YearsSinceLastPromotion (per year)** | 1.09 | 1.01 to 1.19 |
| **BusinessTravel: Travel_Rarely vs Travel_Frequently** | 0.54 | 0.34 to 0.87 |
| **BusinessTravel: Non-Travel vs Travel_Frequently** | 0.22 | 0.09 to 0.53 |
| **JobInvolvement: level 2 vs level 1** | 0.44 | 0.20 to 0.95 |
| **JobInvolvement: level 3 vs level 1** | 0.27 | 0.13 to 0.55 |
| **JobInvolvement: level 4 vs level 1** | 0.15 | 0.05 to 0.41 |
| **EnvironmentSatisfaction: levels 2 and 3 (merged) vs level 1** | 0.32 | 0.20 to 0.52 |
| **EnvironmentSatisfaction: level 4 vs level 1** | 0.22 | 0.13 to 0.39 |
| **JobSat_High (level 4 vs levels 1 to 3)** | 0.49 | 0.32 to 0.77 |
| **StockOptionLevel: levels 1 and 2 (merged) vs level 0** | 0.28 | 0.18 to 0.43 |
| **StockOptionLevel: level 3 vs level 0** | 0.55 | 0.26 to 1.18 (not significant) |
| **JobLevel: level 2 vs level 1** | 0.18 | 0.08 to 0.43 |
| **JobLevel: level 3 vs level 1** | 0.37 | 0.13 to 1.03 (not significant) |
| **JobLevel: levels 4 and 5 (merged) vs level 1** | 0.24 | 0.05 to 1.07 (not significant) |
| **JobRole: HR / Sales Representative (merged) vs Sales Executive** | 0.63 | 0.24 to 1.66 (not significant) |
| **JobRole: Laboratory Technician vs Sales Executive** | 0.45 | 0.18 to 1.17 (not significant) |
| **JobRole: Manager / Director (merged) vs Sales Executive** | 0.11 | 0.03 to 0.41 |
| **JobRole: Manufacturing Director vs Sales Executive** | 0.31 | 0.13 to 0.79 |
| **JobRole: Research Scientist vs Sales Executive** | 0.17 | 0.06 to 0.46 |
| **JobRole: Healthcare Representative vs Sales Executive** | 0.17 | 0.06 to 0.52 |

Reference categories: JobRole = Sales Executive, BusinessTravel = Travel_Frequently, StockOptionLevel = 0, JobLevel = 1, JobInvolvement = 1, EnvironmentSatisfaction = 1. JobRole_Final is significant overall (Wald = 38.5, $p < .001$). Compared with Sales Executives, the odds of leaving are significantly lower for Manager/Director, Research Scientist, Healthcare Representative and Manufacturing Director. The merged HR/Sales Representative group and Laboratory Technicians do not differ significantly from Sales Executives once the other variables are held fixed.

**Overtime came out as the strongest predictor**, in line with what the first chart showed. One thing took me a while to understand: an odds ratio of about 5 does **not** mean overtime workers are 5 times as likely to leave. Odds and probability are different things, so it means the *odds* of leaving are about 5 times higher, with the other variables held fixed.

**Omnibus Tests of Model Coefficients**

| | | Chi-square | df | Sig. |
|---|---|---|---|---|
| Step 1 | Step | 276.871 | 22 | .000 |
| | Block | 276.871 | 22 | .000 |
| | Model | 276.871 | 22 | .000 |

**Model Summary**

| Step | -2 Log likelihood | Cox & Snell R Square | Nagelkerke R Square |
|---|---|---|---|
| 1 | 663.148[a] | .230 | .391 |

a. Estimation terminated at iteration number 6 because parameter estimates changed by less than .001.

**Hosmer and Lemeshow Test**

| Step | Chi-square | df | Sig. |
|---|---|---|---|
| 1 | 29.742 | 8 | .000 |

**Contingency Table for Hosmer and Lemeshow Test**

| | | Attrition = No | | Attrition = Yes | | |
|---|---|---|---|---|---|---|
| | | Observed | Expected | Observed | Expected | Total |
| Step 1 | 1 | 105 | 105.524 | 1 | .476 | 106 |
| | 2 | 99 | 104.626 | 7 | 1.374 | 106 |
| | 3 | 104 | 103.412 | 2 | 2.588 | 106 |
| | 4 | 100 | 100.724 | 5 | 4.276 | 105 |
| | 5 | 100 | 99.446 | 6 | 6.554 | 106 |
| | 6 | 99 | 96.201 | 7 | 9.799 | 106 |
| | 7 | 92 | 90.556 | 14 | 15.444 | 106 |
| | 8 | 89 | 82.380 | 17 | 23.620 | 106 |
| | 9 | 69 | 67.235 | 37 | 38.765 | 106 |
| | 10 | 31 | 37.896 | 76 | 69.104 | 107 |

**Classification Table[c]**

| | | | Predicted | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | Selected Cases[a] | | | Unselected Cases[b] | | |
| | | | Attrition | | | Attrition | | |
| Observed | | | No | Yes | Percentage Correct | No | Yes | Percentage Correct |
| Step 1 | Attrition | No | 866 | 22 | 97.5 | 333 | 12 | 96.5 |
| | | Yes | 104 | 68 | 39.5 | 48 | 17 | 26.2 |
| | Overall Percentage | | | | 88.1 | | | 85.4 |

***Figure 27:*** *Final model: omnibus test, model summary, Hosmer-Lemeshow and classification at cutoff 0.50*

**Variables in the Equation**

| | | B | S.E. | Wald | df | Sig. | Exp(B) | 95.0% C.I.for EXP(B) Lower | Upper |
|---|---|---|---|---|---|---|---|---|---|
| Step 1[a] | OverTime | 1.607 | .210 | 58.612 | 1 | .000 | 4.987 | 3.305 | 7.525 |
| | BusinessTravelValue | | | 13.122 | 2 | .001 | | | |
| | BusinessTravelValue(1) | -.610 | .238 | 6.576 | 1 | .010 | .543 | .341 | .866 |
| | BusinessTravelValue(2) | -1.520 | .452 | 11.334 | 1 | .001 | .219 | .090 | .530 |
| | JobLevel_Final | | | 17.397 | 3 | .001 | | | |
| | JobLevel_Final(1) | -1.704 | .441 | 14.932 | 1 | .000 | .182 | .077 | .432 |
| | JobLevel_Final(2) | -1.001 | .526 | 3.620 | 1 | .057 | .368 | .131 | 1.031 |
| | JobLevel_Final(3) | -1.450 | .774 | 3.509 | 1 | .061 | .235 | .052 | 1.069 |
| | JobRole_Final | | | 38.547 | 6 | .000 | | | |
| | JobRole_Final(1) | -.460 | .494 | .867 | 1 | .352 | .631 | .240 | 1.662 |
| | JobRole_Final(2) | -.792 | .483 | 2.686 | 1 | .101 | .453 | .176 | 1.168 |
| | JobRole_Final(3) | -2.216 | .678 | 10.691 | 1 | .001 | .109 | .029 | .412 |
| | JobRole_Final(4) | -1.159 | .470 | 6.081 | 1 | .014 | .314 | .125 | .788 |
| | JobRole_Final(5) | -1.761 | .505 | 12.148 | 1 | .000 | .172 | .064 | .463 |
| | JobRole_Final(6) | -1.783 | .572 | 9.701 | 1 | .002 | .168 | .055 | .516 |
| | EnvironmentSatisfaction_Final | | | 31.652 | 2 | .000 | | | |
| | EnvironmentSatisfaction_Final(1) | -1.126 | .243 | 21.447 | 1 | .000 | .324 | .201 | .522 |
| | EnvironmentSatisfaction_Final(2) | -1.511 | .288 | 27.429 | 1 | .000 | .221 | .125 | .389 |
| | StockOptionLevel_Final | | | 34.875 | 2 | .000 | | | |
| | StockOptionLevel_Final(1) | -1.278 | .217 | 34.795 | 1 | .000 | .279 | .182 | .426 |
| | StockOptionLevel_Final(2) | -.591 | .386 | 2.337 | 1 | .126 | .554 | .260 | 1.181 |
| | JobSat_Final | -.706 | .229 | 9.534 | 1 | .002 | .494 | .316 | .773 |
| | YearsSinceLastPromotion | .090 | .042 | 4.541 | 1 | .033 | 1.094 | 1.007 | 1.188 |
| | YearsWithCurrManager | -.106 | .039 | 7.362 | 1 | .007 | .899 | .833 | .971 |
| | JobInvolvement | | | 19.389 | 3 | .000 | | | |
| | JobInvolvement(1) | -.824 | .392 | 4.414 | 1 | .036 | .439 | .203 | .946 |
| | JobInvolvement(2) | -1.325 | .371 | 12.748 | 1 | .000 | .266 | .128 | .550 |
| | JobInvolvement(3) | -1.927 | .531 | 13.172 | 1 | .000 | .146 | .051 | .412 |
| | Constant | 3.033 | .682 | 19.768 | 1 | .000 | 20.766 | | |

a. Variable(s) entered on step 1: OverTime, BusinessTravelValue, JobLevel_Final, JobRole_Final, EnvironmentSatisfaction_Final, StockOptionLevel_Final, JobSat_Final, YearsSinceLastPromotion, YearsWithCurrManager, JobInvolvement.

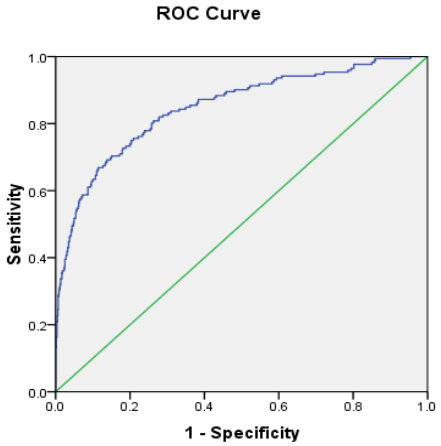
***Figure 28:*** *Final model: variables in the equation (SPSS)*

**Case Processing Summary**

| Attrition | Valid N (listwise) |
|---|---|
| Positive[a] | 172 |
| Negative | 888 |

Larger values of the test result variable(s) indicate stronger evidence for a positive actual state.

a. The positive actual state is Yes.




Diagonal segments are produced by ties.

**Area Under the Curve**

Test Result Variable(s):Predicted probability

| Area | Std. Error[a] | Asymptotic Sig.[b] | Asymptotic 95% Confidence Interval Lower Bound | Upper Bound |
|---|---|---|---|---|
| .848 | .018 | .000 | .814 | .883 |

***Figure 29:*** *ROC curve, training set (AUC 0.848)*

# Chapter 5: The plot twist

At this point I got a test AUC of 0.900 and felt proud. Then I looked at the test output again. It had its own coefficients and its own model fit, and I realized I had refitted the model on the test employees instead of scoring them with the model trained on the 70%. The 0.900 only showed how well a model fits the data it was built on, which is not a test.

I redid it properly: fit on the training set only, then score the test employees. The honest numbers:

| | Training | Test |
|---|---|---|
| **AUC** | 0.848 (95% CI 0.814 to 0.883) | 0.846 (95% CI 0.793 to 0.900) |

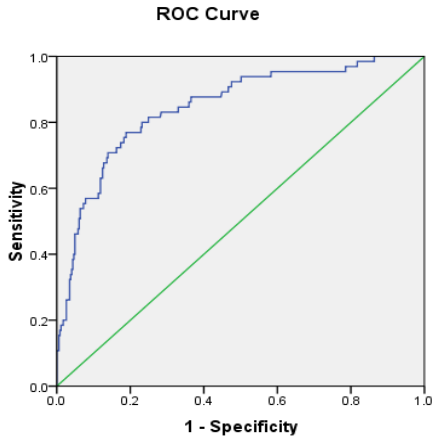
The test AUC is lower than my first result, but it is real, and it almost matches the training AUC, so the model is not just memorizing. An AUC of 0.85 means that if I pick one person who left and one who stayed, the model gives the leaver the higher risk score about 85% of the time.

**Case Processing Summary**

| Attrition | Valid N (listwise) |
|---|---|
| Positive[a] | 65 |
| Negative | 345 |

Larger values of the test result variable(s) indicate stronger evidence for a positive actual state.

a. The positive actual state is Yes.




Diagonal segments are produced by ties.

**Area Under the Curve**

Test Result Variable(s):Predicted probability

| Area | Std. Error[a] | Asymptotic Sig.[b] | Asymptotic 95% Confidence Interval Lower Bound | Upper Bound |
|---|---|---|---|---|
| .846 | .027 | .000 | .793 | .900 |

***Figure 30:*** *ROC curve, test set (AUC 0.846)*

# Chapter 6: From a score to decision

A model gives each person a risk probability, but someone has to decide who to flag. At the default cutoff of 0.50 the model caught only 26.2% of the leavers on the test set, because with only 16% leavers few people reach 50% risk.

So I chose a better cutoff using **the training set only** (Youden's J, which balances catching leavers and avoiding false alarms equally): **0.268**. Then I checked it on the test set:

| At cutoff 0.268 | Training | Test |
|---|---|---|
| **Sensitivity (leavers caught)** | 66.9% (115/172) | 58.5% (38/65), 95% CI about 46% to 70% |
| **Specificity (stayers correctly cleared)** | 88.3% (784/888) | 88.1% (304/345) |
| **Precision (flagged people who really leave)** | 52.5% | 48.1% (38/79), about 3 times the base rate of 15.9% |

**Case Processing Summary**

| | Cases | | | | | |
|---|---|---|---|---|---|---|
| | Valid | | Missing | | Total | |
| | N | Percent | N | Percent | N | Percent |
| Attrition * pred | 1060 | 100.0% | 0 | .0% | 1060 | 100.0% |

**Attrition * pred Crosstabulation**

| | | | pred | | |
|---|---|---|---|---|---|
| | | | 0 | 1 | Total |
| Attrition | No | Count | 784 | 104 | 888 |
| | | % within Attrition | 88.3% | 11.7% | 100.0% |
| | | % within pred | 93.2% | 47.5% | 83.8% |
| | Yes | Count | 57 | 115 | 172 |
| | | % within Attrition | 33.1% | 66.9% | 100.0% |
| | | % within pred | 6.8% | 52.5% | 16.2% |
| Total | | Count | 841 | 219 | 1060 |
| | | % within Attrition | 79.3% | 20.7% | 100.0% |
| | | % within pred | 100.0% | 100.0% | 100.0% |

***Figure 31:*** *Crosstab at cutoff 0.268, training set*

**Case Processing Summary**

| | Cases | | | | | |
|---|---|---|---|---|---|---|
| | Valid | | Missing | | Total | |
| | N | Percent | N | Percent | N | Percent |
| Attrition * pred | 410 | 100.0% | 0 | .0% | 410 | 100.0% |

**Attrition * pred Crosstabulation**

| | | | pred | | |
|---|---|---|---|---|---|
| | | | 0 | 1 | Total |
| Attrition | No | Count | 304 | 41 | 345 |
| | | % within Attrition | 88.1% | 11.9% | 100.0% |
| | | % within pred | 91.8% | 51.9% | 84.1% |
| | Yes | Count | 27 | 38 | 65 |
| | | % within Attrition | 41.5% | 58.5% | 100.0% |
| | | % within pred | 8.2% | 48.1% | 15.9% |
| Total | | Count | 331 | 79 | 410 |
| | | % within Attrition | 80.7% | 19.3% | 100.0% |
| | | % within pred | 100.0% | 100.0% | 100.0% |

***Figure 32:*** *Crosstab at cutoff 0.268, test set*

In plain words, on new employees the model flags about 1 in 5 people and catches about 6 in 10 of those who leave. Nearly half of the people it flags really leave. Accuracy (83.4%) is slightly below "predict everyone stays" (84.1%), which is why I don't use accuracy for a problem where only 16% of people leave.

# Chapter 7: What I am not sure about

I want to be clear about the weak spots:

- **Few leavers per variable:** 172 training leavers for 22 parameters is 7.8 per parameter, below the usual guideline of 10 to 15, so some estimates are imprecise.
- **A small leak:** I chose the variables using all 1,470 rows before splitting, so the test results are slightly optimistic. (The JobRole and JobLevel merges were also decided from a model fitted on all rows; the later merges, the dropped variables and the cutoff used the training set only.)
- **Calibration:** the Hosmer-Lemeshow test fails on the training set ($p < .001$). Most of it comes from one low-risk group with 7 leavers instead of the 1.4 expected, and the top-risk group is slightly under-predicted (76 vs 69.1). The model ranks risk well, but its probabilities are not perfect.
- **A small test set:** with only 65 test leavers, a single split is unstable. The AUC could plausibly be anywhere from 0.79 to 0.90.

## What comes next

- 5-fold cross-validation for a more stable estimate.
- A calibration plot, and trying an interaction such as OverTime × JobLevel.
- Reproducing everything in Python, and comparing with a random forest using SHAP.
- A Power BI dashboard, and the same crosstabs in SQL.

## What I learned

1. Choose the test by the type of data, and report the effect size with the p-value.
2. A model must be judged on data it has never seen.
3. A simpler model that performs the same is the better model.
4. Being honest about a mistake teaches more than a good-looking number.

I'm learning in public. If you spot a mistake or have advice, I would love to hear it.

# Appendix: more SPSS output

**Case Processing Summary**

| | Cases | | | | | |
|---|---|---|---|---|---|---|
| | Valid | | Missing | | Total | |
| | N | Percent | N | Percent | N | Percent |
| BusinessTravelValue * Attrition | 1470 | 100.0% | 0 | .0% | 1470 | 100.0% |

**BusinessTravelValue * Attrition Crosstabulation**

| | | | Attrition | | |
|---|---|---|---|---|---|
| | | | No | Yes | Total |
| BusinessTravelValue | Travel_Frequently | Count | 208 | 69 | 277 |
| | | Expected Count | 232.3 | 44.7 | 277.0 |
| | Travel_Rarely | Count | 887 | 156 | 1043 |
| | | Expected Count | 874.8 | 168.2 | 1043.0 |
| | Non-Travel | Count | 138 | 12 | 150 |
| | | Expected Count | 125.8 | 24.2 | 150.0 |
| Total | | Count | 1233 | 237 | 1470 |
| | | Expected Count | 1233.0 | 237.0 | 1470.0 |

**Chi-Square Tests**

| | Value | df | Asymp. Sig. (2-sided) |
|---|---|---|---|
| Pearson Chi-Square | 24.182[a] | 2 | .000 |
| Likelihood Ratio | 23.760 | 2 | .000 |
| Linear-by-Linear Association | 23.696 | 1 | .000 |
| N of Valid Cases | 1470 | | |

a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 24.18.

**Symmetric Measures**

| | | Value | Approx. Sig. |
|---|---|---|---|
| Nominal by Nominal | Phi | .128 | .000 |
| | Cramer's V | .128 | .000 |
| N of Valid Cases | | 1470 | |

***Figure A1:*** *Crosstab: business travel by attrition*

**Case Processing Summary**

| | Cases | | | | | |
|---|---|---|---|---|---|---|
| | Valid | | Missing | | Total | |
| | N | Percent | N | Percent | N | Percent |
| Department * Attrition | 1470 | 100.0% | 0 | .0% | 1470 | 100.0% |

**Department * Attrition Crosstabulation**

| | | | Attrition | | |
|---|---|---|---|---|---|
| | | | No | Yes | Total |
| Department | Sales | Count | 354 | 92 | 446 |
| | | Expected Count | 374.1 | 71.9 | 446.0 |
| | Research & Development | Count | 828 | 133 | 961 |
| | | Expected Count | 806.1 | 154.9 | 961.0 |
| | Human Resources | Count | 51 | 12 | 63 |
| | | Expected Count | 52.8 | 10.2 | 63.0 |
| Total | | Count | 1233 | 237 | 1470 |
| | | Expected Count | 1233.0 | 237.0 | 1470.0 |

**Chi-Square Tests**

| | Value | df | Asymp. Sig. (2-sided) |
|---|---|---|---|
| Pearson Chi-Square | 10.796[a] | 2 | .005 |
| Likelihood Ratio | 10.490 | 2 | .005 |
| Linear-by-Linear Association | 6.015 | 1 | .014 |
| N of Valid Cases | 1470 | | |

a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 10.16.

**Symmetric Measures**

| | | Value | Approx. Sig. |
|---|---|---|---|
| Nominal by Nominal | Phi | .086 | .005 |
| | Cramer's V | .086 | .005 |
| N of Valid Cases | | 1470 | |

***Figure A2:*** *Crosstab: department by attrition*

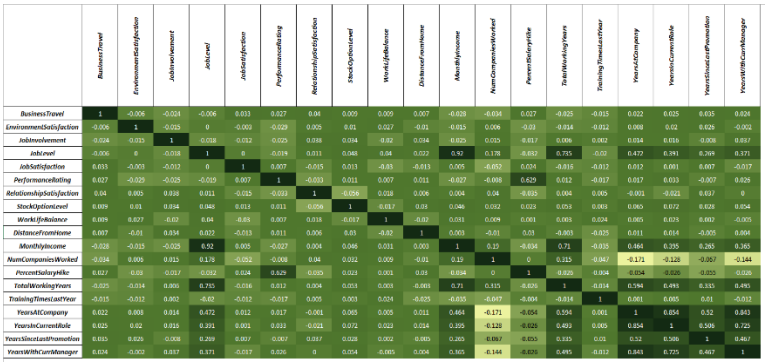


***Figure A3:*** *Spearman correlations: all scale and ordinal variables*

**Categorical Variables Codings**

| | | | Parameter coding | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | Frequency | (1) | (2) | (3) | (4) | (5) | (6) |
| Job Role Final Version | Sales Executive | 235 | .000 | .000 | .000 | .000 | .000 | .000 |
| | HR/Sales Rep | 104 | 1.000 | .000 | .000 | .000 | .000 | .000 |
| | Laboratory Technician | 187 | .000 | 1.000 | .000 | .000 | .000 | .000 |
| | Manager/Director | 121 | .000 | .000 | 1.000 | .000 | .000 | .000 |
| | Manufacturing Director | 103 | .000 | .000 | .000 | 1.000 | .000 | .000 |
| | Research Scientist | 216 | .000 | .000 | .000 | .000 | 1.000 | .000 |
| | Healthcare Representative | 94 | .000 | .000 | .000 | .000 | .000 | 1.000 |
| Job Level Version 2 | 1 | 398 | .000 | .000 | .000 | | | |
| | 2 | 387 | 1.000 | .000 | .000 | | | |
| | 3 | 156 | .000 | 1.000 | .000 | | | |
| | 4 | 119 | .000 | .000 | 1.000 | | | |
| JobInvolvement | 1 | 59 | .000 | .000 | .000 | | | |
| | 2 | 264 | 1.000 | .000 | .000 | | | |
| | 3 | 632 | .000 | 1.000 | .000 | | | |
| | 4 | 105 | .000 | .000 | 1.000 | | | |
| Environment Satisfaction Final Version | 1 | 205 | .000 | .000 | | | | |
| | 2 | 543 | 1.000 | .000 | | | | |
| | 4 | 312 | .000 | 1.000 | | | | |
| Stock Option Level Final Version | 0 | 452 | .000 | .000 | | | | |
| | 1 | 539 | 1.000 | .000 | | | | |
| | 3 | 69 | .000 | 1.000 | | | | |
| BusinessTravelValue | Travel_Frequently | 199 | .000 | .000 | | | | |
| | Travel_Rarely | 745 | 1.000 | .000 | | | | |
| | Non-Travel | 116 | .000 | 1.000 | | | | |

***Figure: A4:*** *Categorical variable codings (training set, n = 1,060). The reference category of each variable is the row where all parameter codings are 0.*

## Frequency

**JobSat Final Verion**

| | | Frequency | Percent | Valid Percent | Cumulative Percent |
|---|---|---|---|---|---|
| Valid | 0 | 1011 | 68.8 | 68.8 | 68.8 |
| | 1 | 459 | 31.2 | 31.2 | 100.0 |
| | Total | 1470 | 100.0 | 100.0 | |

**Stock Option Level Final Version**

| | | Frequency | Percent | Valid Percent | Cumulative Percent |
|---|---|---|---|---|---|
| Valid | 0 | 631 | 42.9 | 42.9 | 42.9 |
| | 1 | 754 | 51.3 | 51.3 | 94.2 |
| | 3 | 85 | 5.8 | 5.8 | 100.0 |
| | Total | 1470 | 100.0 | 100.0 | |

**Environment Satisfaction Final Version**

| | | Frequency | Percent | Valid Percent | Cumulative Percent |
|---|---|---|---|---|---|
| Valid | 1 | 284 | 19.3 | 19.3 | 19.3 |
| | 2 | 740 | 50.3 | 50.3 | 69.7 |
| | 4 | 446 | 30.3 | 30.3 | 100.0 |
| | Total | 1470 | 100.0 | 100.0 | |

**Job Level Version 2**

| | | Frequency | Percent | Valid Percent | Cumulative Percent |
|---|---|---|---|---|---|
| Valid | 1 | 543 | 36.9 | 36.9 | 36.9 |
| | 2 | 534 | 36.3 | 36.3 | 73.3 |
| | 3 | 218 | 14.8 | 14.8 | 88.1 |
| | 4 | 175 | 11.9 | 11.9 | 100.0 |
| | Total | 1470 | 100.0 | 100.0 | |

**JobInvolvement**

| | | Frequency | Percent | Valid Percent | Cumulative Percent |
|---|---|---|---|---|---|
| Valid | 1 | 83 | 5.6 | 5.6 | 5.6 |
| | 2 | 375 | 25.5 | 25.5 | 31.2 |
| | 3 | 868 | 59.0 | 59.0 | 90.2 |
| | 4 | 144 | 9.8 | 9.8 | 100.0 |
| | Total | 1470 | 100.0 | 100.0 | |

***Figure A5:*** *Frequency tables of the final model variables. JobSat_Final: 1 = level 4, 0 = levels 1 to 3. StockOptionLevel: 0, 1 (original 1 and 2), 3(original 3). EnvironmentSatisfaction: 1, 2 (original 2 and 3), 4. JobLevel: 4 = original 4 and 5.*